

**Input
Image**

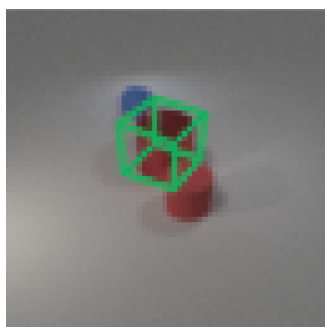
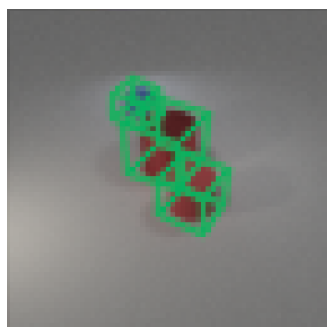


Query

*“find red metal cylinder to the left
behind of red rubber cylinder”*

**Object
Proposals**

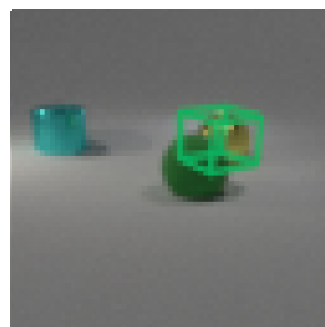
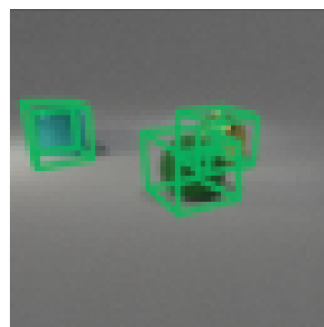
**Detected
object**



*“find yellow metal sphere to the
right behind of green rubber sphere ”*

**Object
Proposals**

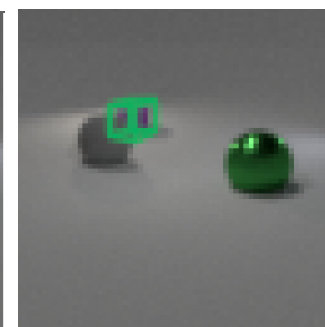
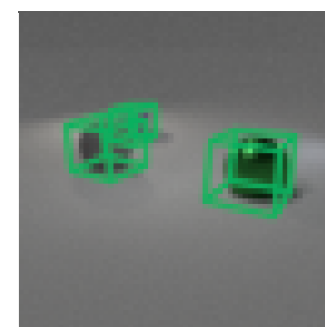
**Detected
object**



*“find purple rubber cube to the
left behind of green metal sphere”*

**Object
Proposals**

**Detected
object**



Prediction

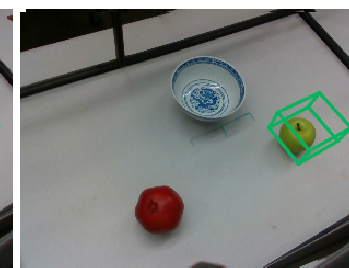
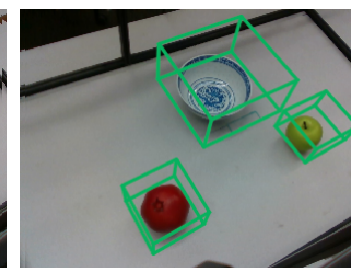
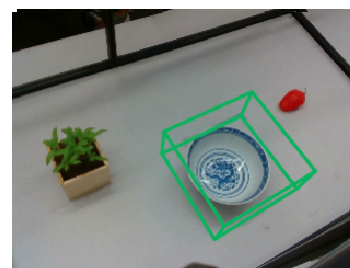
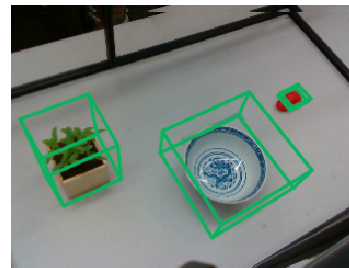
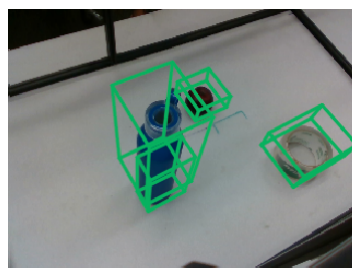
Query

*“find sellotape to the right front
of the plum”*

“find bowl to left front of strawberry ”

*“find green apple to the right of
pomegranate”*

Prediction



(b) Detecting referential spatial expressions. Given a scene and a referential expression, our model localizes the object being referred to in 3D